Math 372: Practice Final Exam #2

Instructions: You have 2 hours. No calculuators, no notes. You must show your work to recieve credit.

Problem 1. A dessert company has 32 distinct desserts on its menu: 6 types of pie, 9 types of cake, and 17 flavors of ice cream. *(You don't need to simplify your answers for this problem)*

(a) After purchasing 4 different desserts, a customer decides he will eat them all in a row, and he cares about the order in which he eats them. How many different ways can he eat his 4 desserts?

(b) Suppose 7 different desserts are randomly selected from the menu. How many ways are there to do this if (i) we care about the order in which the 7 deserts are selected, and (ii) if we don't care about the order?

(c) If 9 desserts are randomly selected, how many ways are there to obtain 3 deserts of each type (pie, cake, and ice cream)?

(d) If 5 desserts are randomly selected, what's the probability that they are all of the same type (i.e., all pie, all cake, or all ice cream)?

Problem 2. Suppose that X is a random variable with probability density function

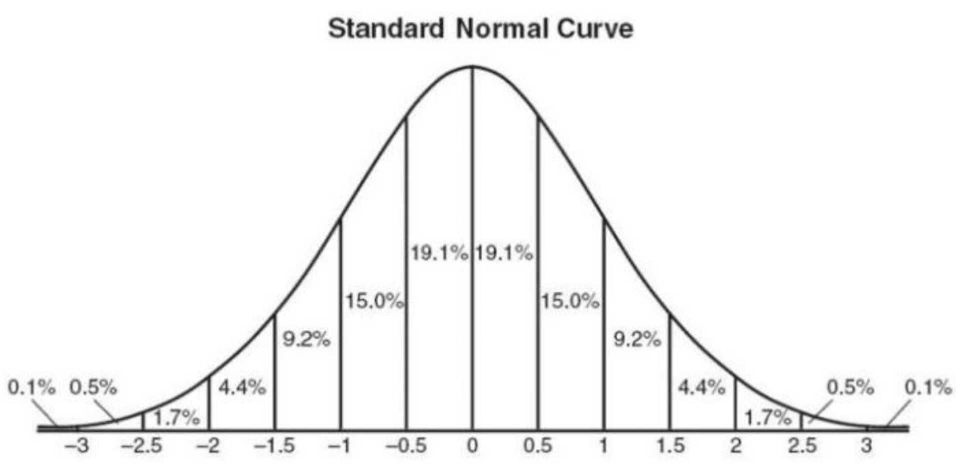
$$f(x) = \begin{cases} \frac{x}{4}e^{-\frac{x}{2}} & : \quad x \geq 0 \\ 0 & : \quad x < 0 \end{cases}$$

(a) Compute $\mathbb{P}\left[X > 3\right]$

(b) Compute $\mathbb{E}\left[X\right]$

Problem 3. A mandolin-making machine in Mordecai's mandolin manufactory makes about 5% defective mandolins even when properly functioning. The mandolins are then packed in crates containing 1900 mandolins each. A crate is examined and found to contain 115 defective mandolins. What is the approximate probability of finding at least this many defective mandolins if the machine is properly adjusted? If you were Mordecai, would hire a technician to check out the machine?

Hint: The following may be useful $\sqrt{1900} = 10\sqrt{19}$. Also, the pdf of a standard normal random variable has graph



Problem 4.

Morkbork, an alien from the star system Dihedral-5, is testing out some cool new *quantum coins* that she recently purchased at the interstellar gas station. The packaging advertised that the atoms in the coins are “entangled,” so that if you flip both coins at the same them, they are more likely to land on the same side. Let H and T denote the events that the first coin lands on “heads” and “tail”. Let $\widetilde{H}$ and $\widetilde{T}$ be the events that the second coin lands “heads” and “tails”. Morkbork is skeptical, but she decides to test the coins by flipping them together 100 times. She records her results in the table:

	$\widetilde{H}$	$\widetilde{T}$
H	30	10
T	20	40

Morkbork doesn’t know if the coins are “fair” or not, and she doesn’t care. She only wants to know if they are “entangled” or “independent”. So naturally she decides to do a chi-squared test of independence.

(a) What is the null hypothesis?

(b) From the table, estimate the probabilities $\mathbb{P}[H], \mathbb{P}[\widetilde{H}], \mathbb{P}[T]$, and $\mathbb{P}[\widetilde{T}]$.

(c) Assuming the null hypothesis is true, estimate the probabilities

$$\mathbb{P}[H \cap \widetilde{H}], \quad \mathbb{P}[H \cap \widetilde{T}], \quad \mathbb{P}[T \cap \widetilde{H}], \quad \text{and} \quad \mathbb{P}[T \cap \widetilde{T}].$$

(d) Based on your answer ot the previous part, make a table of estimated expected counts, assuming H_0.

(e) Compute the χ^2 statistic.

(f) Circle the correct phrase to complete the sentence: *A larger chi-squared value means the observed data is (very different from / very similar to) the expected data.*

(g) Fill in the blank: *In this problem, the chi-squared statistic χ^2 has a chi-squared distribution with ______ degrees of freedom.*

(h) Morkbork’s five-year old brother, Glorbmorlk, asks her what “p-value” means. Help Morkbork give a simple explanation to Glorbmorlk by describing in several compete sentences what p-value means.

(i) Set up, but do not evaluate, an integral to compute a p-value based on your χ^2 statistic. Hint: the density to the chi-squared statistic is

$$f(x) = \begin{cases} \frac{1}{\sqrt{2\pi}} x^{-1/2} e^{-x/2} & : \quad x \geq 0 \\ 0 & : \quad x < 0 \end{cases}$$

(j) The p-value turns out to be a very small number, about 3×10^{-63}. How should Morkbork interpret this result?

Problem 5. Let T be the waiting time, in hours, between successive purchases at a furniture consignment store. Suppose that T has pdf

$$f_T(t) = Ce^{-\frac{t}{3}}\mathbf{1}_{[t\geq 0]} = \begin{cases} Ce^{-\frac{t}{3}} & : \ t \geq 0 \\ 0 & : \ t < 0 \end{cases}$$

where $C > 0$ is some number that you'll have to determine.

(a) Find a value of C so that $f_T(t)$ is a valid pdf.

(b) What is the probability that the time between purchases is greater than 6 minutes?

(c) Compute $\mathbb{E}[T]$.

(d) Compute $\mathrm{Var}(T)$. *Hint: integraton by parts*

Problem 6. Consider the following three data points, of the form (x, y):

$$(-1, 4), \quad (0, 5), \quad \text{and} \quad (1, 9)$$

(a) Plot the data points on the xy-plane

(b) Suppose we have a line $y = C + Dx$. Let $\vec{e} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$ be the vector whose entries are the vertical distances from the data points to the line. Write out $\vec{e}$.

(c) Find the values of C and D which minimize the length of $\|\vec{e}\|^2 = e_1^2 + e_2^2 + e_3^2$.

(d) Sketch the line $y = C + Dx$ using your values of C and D.

Problem 7. You are dealt a 5-card poker hand.

(a) What is the probability that the fifth card you recieve is a spade?

(b) What is the probability that your hand contains at least one spade?

(c) What is the conditional probability that the fifth card you recieve is a spade, given that your hand contains at least one spade?

(d) What is the probability that you receive your first spade on the third card dealt?

(e) Find the expected number of suits appearing in your hand. (Hint: let S be the event that you have at least one spade, and define D, C, H similarly for diamonds, clubs, and hearts. The number of suits in your hand is $N = \mathbf{1}_S + \mathbf{1}_D + \mathbf{1}_C + \mathbf{1}_H$.)